

RECSYS CHALLENGE 2026 · MUSIC-CRS

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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11

retrieval branches

146

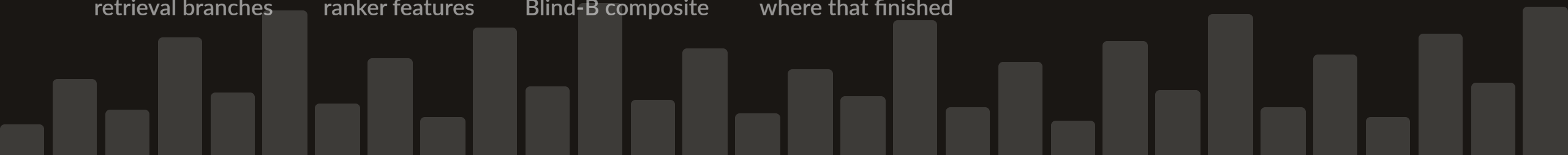
ranker features

0.3811

Blind-B composite

29/40

where that finished





The short version

SUMMARY

CORE

WHAT WE BUILT

An LLM reads each turn into a **typed state**; eleven retrieval branches search one index; a **LightGBM** ranker picks 20; an LLM writes the reply.

HOW IT SCORED

0.3811

Blind-B · 29th of 40
nDCG@20 fell from 0.438 on Blind-A to 0.254.

WHAT WE FOUND

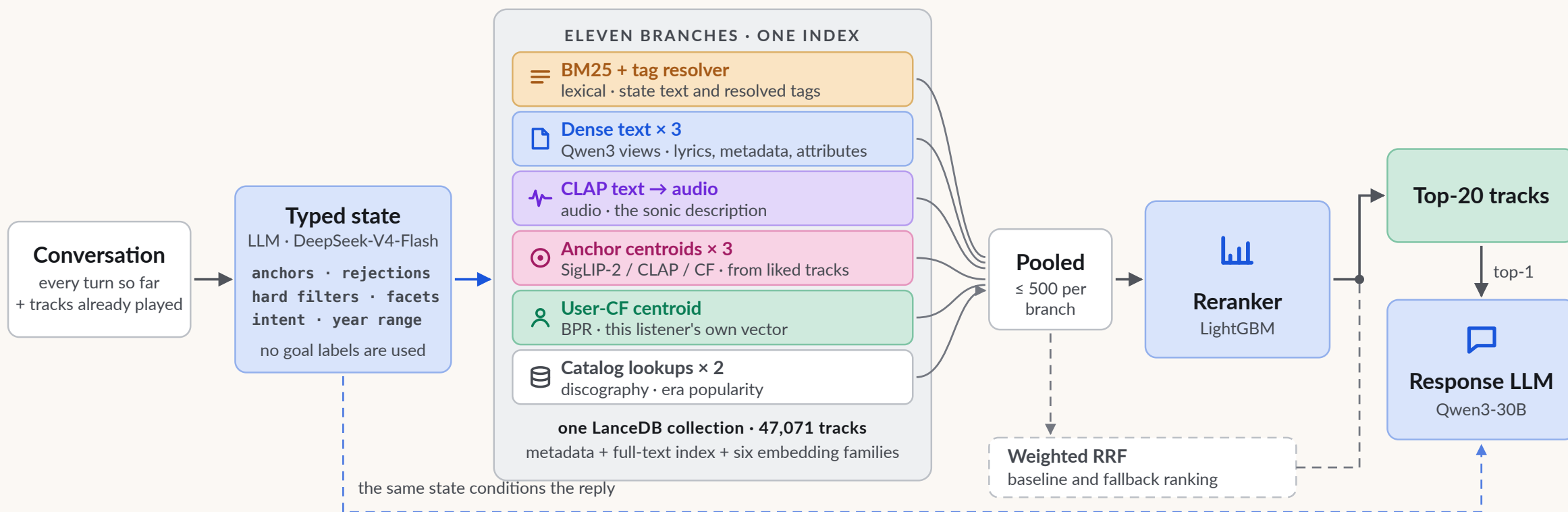
31 of 43

diagnosed failures were ranking

Extraction was **3**. The state was richer than what the ranker used.

The system understood the conversation more often than it **acted on it**.

One state, eleven branches, one learned ranker





State extraction on a real Blind-B turn

Earlier turns: we played Neko Case's "Calling Cards" ("not quite the one"), then "Man" ("not it either").

User, now: "...much more subdued... a specific sense of place in the lyrics... a somewhat stark, almost a cappella delivery."

Extracted state (abridged):

request_type: hidden_target

anchor artist: Neko Case (must_use)

rejected tracks: "Calling Cards", "Man"

facets: mood=subdued; lyrical_theme=sense of place;
sonic=stark almost a cappella

Top-1: Neko Case – "Bracing For Sunday"

WHERE EACH FIELD GOES

- The **anchor** drives the centroid searches
- The **facets** become text and audio queries
- The **rejections** filter the fused pool

AND WHERE THEY DO NOT

Nothing here becomes a hard filter on album, and the year window never reaches the index. That gap is the main diagnostic later.



What the ranker leans on

FEATURE FAMILY	#	GAIN
Bi-encoder cosine	1	59.0%
Per-branch rank and score	66	11.5%
Session and artist affinity	18	10.7%
Popularity · state · tags	31	9.2%
Other similarity cosines	17	8.9%
Constraints and rejections	13	0.7%

SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.



How it scored

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final, 29th of 40	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

category and lexical barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



Asked for, served

WENT WRONG

CORE

1

*“the Kamelot track from **Silverthorn**”*

Kamelot — Fallen Star

from *Haven*. We extracted “Silverthorn” but never filtered on it.

CONSTRAINT

2

*“exactly **126.70 bpm**, in **G minor**”*

no tempo or key in the catalog

and we added no data from outside it

NO DATA

3

*“**‘Watercolors’** by **Pat Metheny**”*

Pat Metheny — Alfie

reply: “Got it — here’s ‘Alfie’...” — “Watercolors” is not in the catalog

RESPONSE



Where the failures were — all 80 Blind-B turns

ANALYSIS CORE

DIAGNOSED GAP	TURNS	MEANING
Ranking	31	a better, non-violating candidate sat below our top-20
Constraint not applied	8	an extracted filter never reached the pool
State extraction	3	the extractor missed something the user said
Entity resolution	1	a name never resolved to a catalog ID

TOP-1 FLAGGED

17 / 80

the track we led with violated something the state had recorded

CATEGORIES OVERLAP

43 diagnoses across 32 distinct turns. This is a heuristic audit, not a relevance judgment — Blind-B exposes no labels.

Extraction is **3 of 43** diagnoses. The failure is overwhelmingly downstream of understanding the conversation.



Extracted, then ignored

WENT WRONG

CORE



*"I gotta break away from **Deltron** and **Del** completely"*

Dead Prez — Hip Hop

state held both artists as avoid

IGNORED



*"branching out to **other** singer-songwriter artists"*

Ryan Adams — When The Stars Go Blue

state held Ryan Adams as avoid

IGNORED



*"still focusing on **Juanes**... haven't found the depth I'm after"*

Juanes — Me Enamorado De Ti

state held Juanes as rejected

IGNORED

Three sessions, one failure: the listener asked to move on, the state recorded it, and **the ranker served that artist anyway.**



What the training labels look like

WENT WRONG

CORE



*"I keep asking for **Arcade Fire** — that's the only song I want"*

label: **Arctic Monkeys — This House is a Circus**

DOES NOT



*"similar chill vibe, but **from a different artist**"*

label: **Bonobo — Jets**

MOVES



*"what **other artists** fit this darkwave style?"*

label: **Tropic of Cancer — The Dull Age**

MOVES



*"what **other** orchestral tracks would you recommend?"*

label: **Nick Phoenix — Kogan**

MOVES

Three of four are annotated **MOVES TOWARD GOAL** — the dataset records them as having helped. Train on this and the model learns to anchor. **Ours does.**



What we would do differently

CONCLUSION

CORE

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.



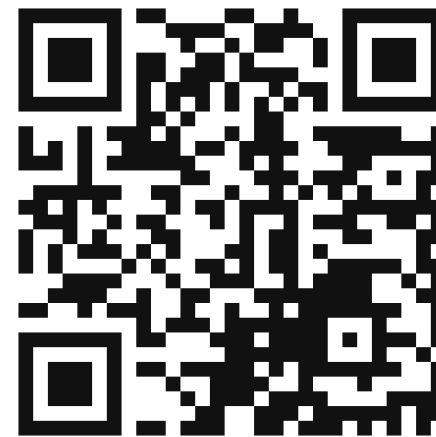
Everything is released

IN THE REPO

- Pipeline code, configs, trained models
- Cached per-turn extraction states
- The 113,393 re-judged labels
- A frozen-LLM-cache replay bundle

ZERO-CREDENTIAL REPLAY

Every LLM call is served from a curated frozen cache that **hard-fails on a miss** rather than falling through to a paid call. Both submissions reproduce without any API key.



Paper, poster, slides, code, data

npatta01.github.io/music-crs-2026